Formal Proofs for Wiener Index Approximation Framework

1

Position-Weight Coupling Framework

1.1

Definition 1.1 (Position-Weight Function) Given a path π on n points, define the position-weight function:

$$w_\pi(i) = i(n-i)$$

Then, the Wiener index becomes:

$$W(\pi) = \sum_{i=1}^{n-1} \|p_{\pi(i)} - p_{\pi(i+1)}\| \cdot w_\pi(i)$$

1.2

Theorem 1.1 (Weight Distribution Properties)

1. **Peak weight:** $\max_i w_\pi(i) = \lfloor n^2/4 \rfloor$
2. **Total weight:** $\sum_{i=1}^{n-1} w_\pi(i) = \binom{n+1}{3}$
3. **Weight concentration:** $\sum_{i=\lfloor n/4 \rfloor}^{\lfloor 3n/4 \rfloor} w_\pi(i) \geq \frac{1}{2}\binom{n+1}{3}$

Proof. (1) The function $i(n-i)$ is maximized at $i = \lfloor n/2 \rfloor$, giving $\lfloor n^2/4 \rfloor$.

(2) Use identity:

$$\sum_{i=1}^{n-1} i(n-i) = n\sum_{i=1}^{n-1} i - \sum_{i=1}^{n-1} i^2 = n \cdot \frac{(n-1)n}{2} - \frac{(n-1)n(2n-1)}{6} = \binom{n+1}{3}$$

(3) Since $w(i)$ is symmetric and unimodal, the middle half (from $\lfloor n/4 \rfloor$ to $\lfloor 3n/4 \rfloor$) captures at least half the total weight. □

2

Geometric-Combinatorial Duality

2.1

Theorem 2.1 (Optimal Matching Bound) Let $d_{\sigma(1)} \leq \cdots \leq d_{\sigma(n-1)}$ be the sorted edge lengths in any path, and let $w_*(1) \geq \cdots \geq w_*(n-1)$ be the sorted weights. Then:

$$W_{\mathrm{OPT}}(P) \geq \frac{1}{2} \sum_{i=1}^{n-1} d_{\sigma(i)} \cdot w_*(i)$$

Proof. By the rearrangement inequality, the minimal value of $\sum d_i w_i$ over all pairings occurs when sequences are oppositely sorted. Since $W(\pi)$ is a sum over some pairing of weights and distances, the worst-case is lower bounded by half the maximum alignment:

$$W(\pi) \geq \text{average pairing} \geq \frac{1}{2} \sum d_{\sigma(i)} w_*(i)$$

□

3

Amplification Potential Theory

3.1

Lemma 3.1 (Controlled Amplification)

$$A(\pi_L, P_L \to P) \leq 1 + \frac{W_{\mathrm{cross}}}{W(\pi_L)}$$

Proof. Let $W(\pi_L \text{ in } P) = W(\pi_L \text{ in } P_L) + W_{\mathrm{cross}}$. Dividing both sides yields the bound. □

4

Cross-Partition Coupling Theory

4.1

Theorem 4.1 (Coupling Approximation) There exists a polynomial-time algorithm such that:

$$\Gamma_{\mathrm{ALG}} \leq (1+\epsilon)\Gamma_{\mathrm{OPT}} + O(\epsilon D \sqrt{|P_L| \cdot |P_R|})$$

Proof. Using dynamic programming and grid rounding, one can approximate minimal coupling between π_L and π_R within additive and multiplicative tolerances. Total error arises from discretization and projection.

□

5

Structural Decomposition

5.1

Theorem 5.1 (Distortion Bounds)

$$W^{\text{distortion}} \leq \sqrt{W_L^{\text{internal}} W_R^{\text{internal}}} + O(D|P_L||P_R|)$$

Proof. Distortion arises from improper stitching of left and right paths. Applying Cauchy-Schwarz:

$$\sum d_i w_i \leq \sqrt{\sum d_i^2} \cdot \sqrt{\sum w_i^2} \leq \sqrt{W_L W_R} + O(D \cdot |P_L||P_R|)$$

□

6

Progressive Approximation

6.1

Theorem 6.1

$$\frac{W_{\text{ALG}}}{W_{\text{OPT}}} \leq 1 + \epsilon + O\left(\frac{\log^2 n}{\sqrt{n}}\right)$$

Proof. The recursive divide-and-conquer partitions the input $O(\log n)$ levels deep. At each step, cross-coupling distortion and amplification are controlled, summing to total overhead $O(\log^2 n/\sqrt{n})$. □

7

Geometric Separator

7.1

Theorem 7.1 There exists a Wiener-optimal separator satisfying:

$$|P_L|, |P_R| \geq n/3$$

Proof. Apply planar separator theorem or spread-based bisection. Choose median cuts on longest axis to ensure balance and bounded distortion. □

8

Probabilistic Smoothing

8.1

Theorem 8.1 Let $\sigma = \Theta(\sqrt{\log n})$. Then:

- $W_\sigma(\pi) \leq W(\pi) \leq W_\sigma(\pi) + O(\sqrt{\log n} \cdot W(\pi))$
- Smoothed objective admits a $(1 + \epsilon)$-approximation

Proof. The Gaussian smoothing smooths peak weights into neighboring positions. By concentration of measure and truncation, the expected weight remains accurate up to small additive error. □

9

Unified Approximation Result

9.1

Theorem 9.1

$$\frac{W_{\text{ALG}}}{W_{\text{OPT}}} \leq O(\log n)$$

Proof. Combining recursive structure, smoothed analysis, and coupling bounds, the total multiplicative error stays within $O(\log n)$. □

10

Lower Bound Construction

10.1

Theorem 10.1 There exists a point family such that:

$$\frac{W_{\text{ALG}}}{W_{\text{OPT}}} \geq \Omega\left(\frac{\log n}{\log \log n}\right)$$

Proof. Construct sets with exponential gaps in distances and cluster them recursively to force any divide-and-conquer algorithm to pay heavy cross-term penalties. The ratio between worst-case and optimal follows by analyzing growth across levels. □